This assignment will not be graded and is only for practice.

Level 1

1) Let $f(x,y,z) = (xy, x+y+z^2)$, where $x, y, z \in \mathbb{R}$. Choose the set of correct options. ***1 point***

- [ ] The domain of f is $\mathbb{R}^2$ and the co-domain of f is $\mathbb{R}^2$
- [ ] The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$
- [ ] f is continuous everywhere in its domain
- [ ] f is not continuous anywhere in its domain

No, the answer is incorrect.

Score: 0

Accepted Answers:

The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$

f is continuous everywhere in its domain

2) Let $f(x,y) = \begin{cases} \frac{\cos y \sin x}{x} & x = 0 \\ \cos y & x = 0 \end{cases}$. ***1 point***

Choose the correct options.

- [ ] f is not defined at $(0,0)$
- [ ] f is defined at $(0,0)$
- [ ] f is continuous at $(0,0)$
- [ ] f is not continuous at $(0,0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

f is defined at $(0,0)$

f is continuous at $(0,0)$

3) Let $h(u, v) = \sin(uv), g(x) = x^3, f(y) = \cos y$. Let $p(x, y) = h(g(x), f(y))$ **1 point**

Choose the correct option(s).

- [ ] p is a scalar-valued function
- [ ] p is a vector-valued function
- [ ] p is continuous everywhere since $g.f$ is continuous everywhere in its domain
- [ ] p is continuous everywhere, g is also continuous everywhere while f is not continuous everywhere in its domain

No, the answer is incorrect.
Score: 0

Accepted Answers:

p is a scalar-valued function
p is continuous everywhere since $g.f$ is continuous everywhere in its domain

4) Let $p(t) = (ln(1-t), \frac{1}{t}, 3t)$. **1 point**

Choose the incorrect option(s).

- [ ] p is continuous for all real t less than 1 but not equal to zero
- [ ] p is continuous on $\mathbb{R}$
- [ ] p is continuous in $\{t \in \mathbb{R} \mid t < 1\}$
- [ ] p is continuous in $\{t \in \mathbb{R} \mid t = 1, t = 0\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

p is continuous on $\mathbb{R}$
p is continuous in $\{t \in \mathbb{R} \mid t < 1\}$
p is continuous in $\{t \in \mathbb{R} \mid t = 1, t = 0\}$

5) Let $f(x, y) = ln(x^2 + y^2 - 1)$. Identify where f is continuous. Note: Equation of a circle with radius r and center (a, b) is given by $(x - a)^2 + (y - b)^2 = r^2$ **1 point**

- Everywhere in R^2
- Everywhere in $\mathrm{R}^2 \setminus \{(1, 0), (0, 1)\}$
- Everywhere in R^2 outside the circle with radius 1 and center $(0, 0)$
- None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:

Everywhere in R^2 outside the circle with radius 1 and center $(0, 0)$

Level 2

6) Let $g(u, v, w) = \frac{uv}{w}$ and $f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$ **1 point**

where $P(x, y, z) = e^{x^2+y}$, $Q(x, y, z) = \sqrt{y^2 + z^2 + 3}$, $R(x, y, z) = \sin(xyz) + 5$.

Choose the correct option(s).

- f is a vector-valued function
- f is a scalar-valued function
- f is continuous everywhere in its domain.
- P is continuous everywhere in its domain, while Q is not continuous everywhere in its domain

No, the answer is incorrect.
Score: 0

Accepted Answers:

f is a scalar-valued function
f is continuous everywhere in its domain.

7) Let **1 point**

$$f : \mathrm{R}^2 \to \mathrm{R}, f(x, y) = \{\begin{matrix} \frac{x^2}{x^2+y^2} & (x, y) = (0, 0) \\ 0 & (x, y) = (0, 0) \end{matrix}.$$

Choose the correct option(s).

- [ ] $\lim_{(x,y)\to(0,0)} f$ does not exist
- [ ] $\lim_{(x,y)\to(0,0)} f$ exists
- [ ] f is continuous at $(0, 0)$
- [ ] f is not continuous at $(0, 0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\lim_{(x,y)\to(0,0)} f$ does not exist
f is not continuous at $(0, 0)$

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8) Let **1 point**

$$f : \mathrm{R}^2 \to \mathrm{R}, f(x, y) = \{\begin{matrix} \frac{xy}{x^2+y^2} & (x, y) = (0, 0) \\ 0 & (x, y) = (0, 0) \end{matrix}.$$

Choose the correct option(s).

- [ ] $\lim_{(x,y)\to(0,0)} f$ does not exist
- [ ] $\lim_{(x,y)\to(0,0)} f$ exists
- [ ] f is continuous at $(0, 0)$
- [ ] f is not continuous at $(0, 0)$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\lim_{(x,y)\to(0,0)} f$ does not exist

f is not continuous at $(0, 0)$

9) Let $f(r) = \frac{1}{<r,r>-1}$, where $r = (x, y, z)$ and the inner product $<,>$ is the dot product *1 point*
Choose the set of incorrect options.

- [ ] f is continuous throughout $\mathbb{R}^3$
- [ ] f is continuous throughout $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$
- [ ] f is discontinuous on the unit sphere $\{r \in \mathbb{R}^3 : \|r\| = 1\}$
- [ ] f is continuous throughout $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

f is continuous throughout $\mathbb{R}^3$
f is continuous throughout $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$
f is continuous throughout $\mathbb{R}^3 \setminus \{(1, 0, 0)\}$

10) Let $f : \mathbb{R}^2 \to \mathbb{R}, f(x, y)= \begin{cases} \frac{x^k y}{x^4+y^4} & (x, y) = (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$. Find the minimum possible positive integer value of k such that f is continuous at (0,0).

No, the answer is incorrect.
Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Your score is: 0/10